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(a) amplitudes:

Particles in a progressive wave have the same amplitude.

Particles in a stationary wave have amplitudes ranging from zero at the nodes to maximum amplitude at the antinodes.

phases:

Particles within a wavelength in a progressive wave have different phases.

Particles in a stationary wave between adjacent nodes oscillate with the same phase and particles between adjacent segments oscillate π rad out-of-phase.

Particles at the node do not oscillate.

(b) (i) Wavelength is the distance between two adjacent particles that oscillate in phase.

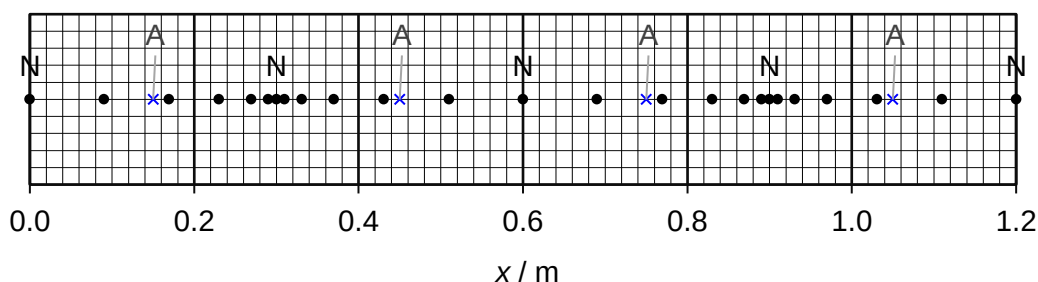
$$2\lambda = 1.20 \text{ m} \quad (2 \text{ d.p.})$$

$$\lambda = 0.600 \text{ m} \quad (3 \text{ s.f.})$$

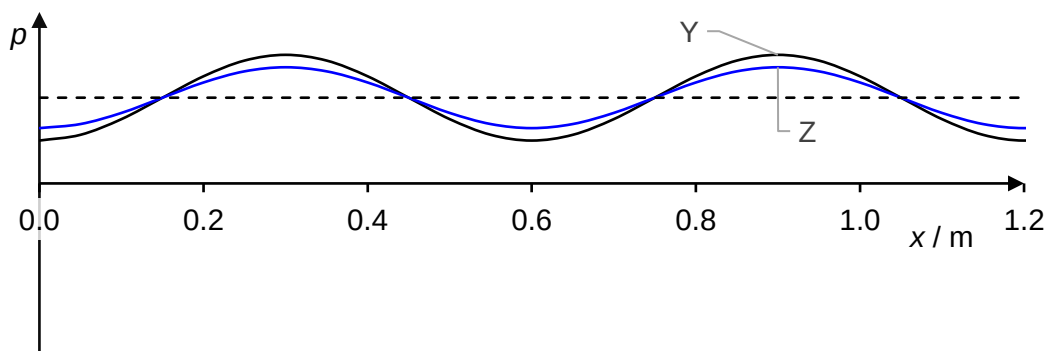
(ii) There are 6 intervals between 0.00 m and 0.30 m ($1/2$ wavelength) indicating that the separation between particles when they are at equilibrium is 0.05 m. The particle with equilibrium position at 0.15 m is at the amplitude position of 0.19 m at t_0 .

$$x_0 = 0.19 - 0.15 = 0.04 \text{ m} \quad (2 \text{ d.p.})$$

(c) (i)



(ii)



1. Pressure at the nodes is highest or lowest compared to the initial atmospheric pressure.

*Correct high and low pressure positions.

2. For stationary waves, energy is not transferred and the waveform does not progress forward. Positions of nodes and antinodes remain unchanged. Only the displacement of the particles between the nodes changes.

*Graph of smaller amplitude at $t_1 + \frac{T}{8}$ compared to at t_1 .

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- (d) (i)** Stationary wave is formed in the air column in the tube when an antinode is at the mouth of the tube and a node is at the water surface.

When first loud note is heard,

$$L_1 + c = \frac{1}{4} \lambda$$

$$c = \frac{1}{4} \lambda - L_1$$

$$= \frac{1}{4} (0.600) - 0.144$$

$$= 0.150 - 0.144$$

$$= 0.006 \text{ m (3 d.p.)}$$

- (ii)** When the next loud note is heard,

$$L_2 + c = \frac{3}{4} \lambda$$

$$L_2 = \frac{3}{4} \lambda - c$$

$$= \frac{3}{4} (0.600) - 0.006$$

$$= 0.444 \text{ m (3 d.p.)}$$

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4. (a) (i) The electric field strength due to each particle is directed towards the left. As the